higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE

CONTROL SYSTEMS N6

(8080016)

3 December 2019 (X-Paper)
09:00–12:00

REQUIREMENTS: Three-cycle semilogarithmic graph paper

This question paper consists of 7 pages, 2 diagram sheets, a formula sheet of 2 pages and 3 Laplace transform tables.

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DEPARTMENT OF HIGHER EDUCATION AND TRAINING
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE
CONTROL SYSTEMS N6
TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Use neat, labelled circuit diagrams in the explanation of answers only where requested.
5. Use only a BLACK or BLUE pen.
6. Use drawing equipment and a pencil for ALL sketches and diagrams.
7. Write neatly and legibly.

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QUESTION 1

Indicate whether the following statements are TRUE or FALSE. Choose the answer and write only 'True' or 'False' next to the question number (1.1–1.10) in the ANSWER BOOK.

1.1 A system can be electrical, mechanical, biological and economical.

1.2 Rise time is the time required for the response to a unit-step function to rise from 10 to 90 percent of its final value.

1.3 In an open-loop system the output has an effect on the input in order to maintain the output at a desired value.

1.4 The gain margin and phase margin are known as the margin of safety and are used to assess the stability of a system.

1.5 Settling time is the time required for the response to a unit-step function to reach and remain within 98 percent of its final value.

1.6 A step input remains at a constant value.

1.7 An open-loop control system does not become unstable due to the absence of feedback elements which may make the output uncontrollable.

1.8 An unstable system will remain at rest unless excited by an external source and will return to rest once all excitations are removed.

1.9 The transient period is a period of change when the output of the system changes from an initial value to a new value.

1.10 In a closed-loop control system the reverse path is the transmission path from the comparator to the controlled output signal.

(10 × 1) **[10]**

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QUESTION 2

2.1 Determine, with the aid of block-diagram algebra reduction, the control ratio for the block diagram in FIGURE 1. (7)

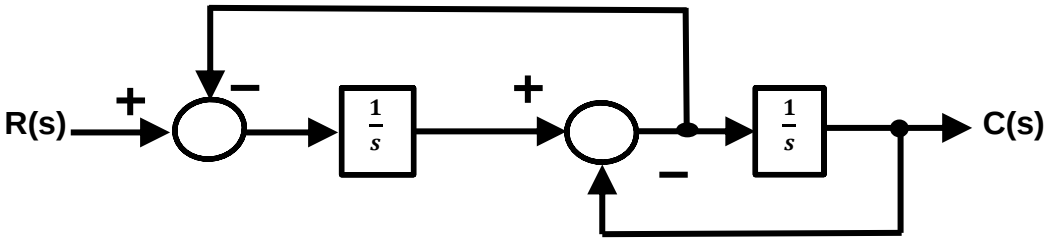


FIGURE 1

2.2 State THREE effects of a disturbance on a control system. (3)

[10]

QUESTION 3

3.1 Plot a point-by-point Bode diagram for the following data on three-cycle semilogarithmic graph paper:

ω (rad/sec)	1	3	6	10	30	300	1 000
Gain (dB)	40	30	25	20	-10	-30	-40
θ (degrees)	-15°	-25°	-35°	-55°	-95°	-115°	-155°

(5)

3.2 Determine the following from the graph:

3.2.1 The gain crossover frequency (1)

3.2.2 The phase crossover frequency (2)

3.2.3 The phase margin (1)

3.2.4 The gain margin (1)

[10]

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QUESTION 4

DIAGRAM SHEET 1 (attached) illustrates a closed-loop gain-versus-phase plot on a Nichols chart.

4.1 Use the plot to tabulate the closed-loop gain and phase values when ω = 5 rad/s, 10 rad/s and 150 rad/s. (6)

4.2 Use the Nichols chart to determine the following:

4.2.1 The undamped resonant frequency

4.2.2 The bandwidth

4.2.3 The peak magnitude gain value

4.2.4 The peak magnitude phase value

(4 × 1) (4)

[10]

QUESTION 5

DIAGRAM SHEET 2 (attached) illustrates a root locus plot of a system as the amplifier gain varies from zero to infinity.

Use the root locus plot to determine the following:

5.1 The open-loop poles and zeroes (2)

5.2 The damped resonant frequency ω_d at a damping ratio of 0,707 (3)

5.3 The undamped resonant frequency 'ω_n at a damping ratio of 0,707 (1)

5.4 The gain constant K_o' at a damping ratio of 0,707 (3)

5.5 The frequency at which the system becomes unstable (1)

[10]

QUESTION 6

6.1 Convert each of the following Laplace transform functions to a function of s:

6.1.1 $F(t) = \frac{10}{49}e^{-7t} + 7t - 1$

6.1.2 $F(t) = 115\left[1 - \frac{(5+t)}{5}\, e^{\frac{-t}{5}}\right]$

(2 × 2) (4)

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6.2 Convert each of the following Laplace transform functions to a function of t:

6.2.1 $F(s) = \frac{30\omega}{(s^2 + \omega^2)(3s + 1)}$

6.2.2 $F(s) = \frac{24}{s^2(s + 4)(s + 3)}$

(2 × 3) (6)
[10]

QUESTION 7

A ramp function of 2 units per time interval is applied to a system with the control ratio:

$$\frac{C(s)}{R(s)} = \frac{12s}{s^2 + 7s + 12}$$

Determine the following:

7.1 The total response of the system (6)

7.2 The initial value of the response (2)

7.3 The final value of the response (2)
[10]

QUESTION 8

8.1 Draw a neat, labelled schematic diagram of a differential capacitive angular displacement transducer. (5)

8.2 Explain the operating principle of a differential capacitive angular displacement transducer. (5)
[10]

QUESTION 9

9.1 List the THREE main functions of hydraulic fluid. (3)

9.2 Differentiate between a *rotary pump* and a *centrifugal pump*. (2)

9.3 Give the advantage of using hydraulic motors. (2)

9.4 Name TWO types of semirotary actuators or motors. (2)

9.5 Explain the term *derivative control* relating to process controllers used in hydraulic systems. (1)
[10]

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QUESTION 10

10.1 Different Lissajous figures are obtained on the screen of an oscilloscope at different phase-shift angles.

Name the figures displayed at each of the following phase-shift angles:

10.1.1 0° phase shift

10.1.2 0°–90° phase shift

10.1.3 90° phase shift

10.1.4 90°–180° phase shift

(4 × 1) (4)

10.2 A sinusoidal response causes a pulse to have an effect on an RC circuit.

Draw the waveforms produced in response to a pulse when:

10.2.1 R.C << T

10.2.2 R.C = T

10.2.3 R.C >> T

(3 × 1) (3)

10.3 Calculate the value of the unknown frequency (f_V) in FIGURE 2.

HINT: $\frac{f_H}{f_V} = \frac{\text{Number of vertical tangencies}}{\text{Number of horizontal tangencies}}$

$f_H = 1\,000\ H_z$

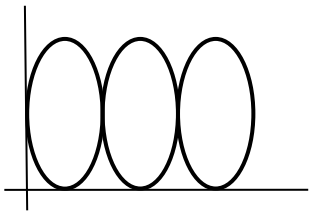


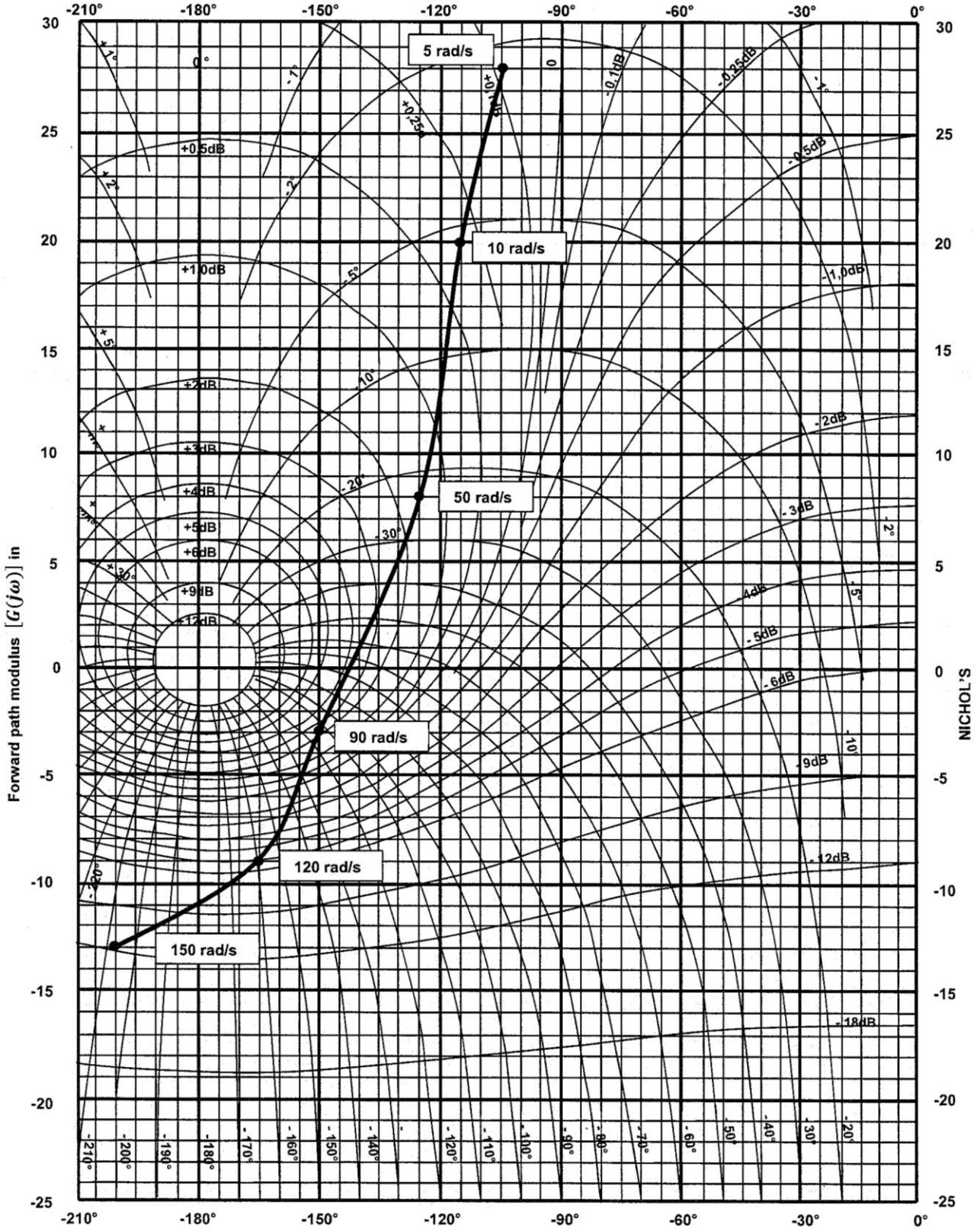
FIGURE 2

(3)
[10]

TOTAL: 100

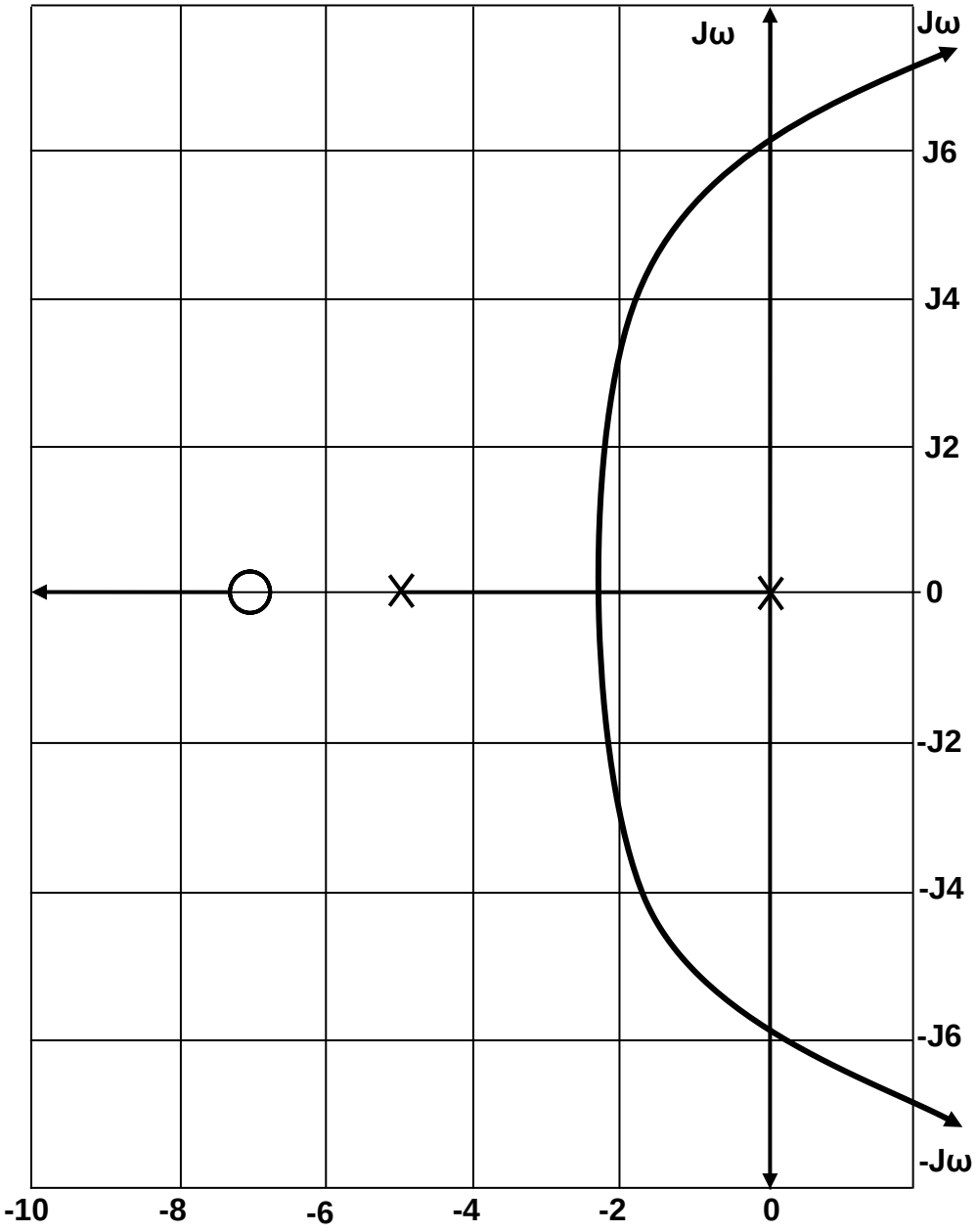
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DIAGRAM SHEET 1



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DIAGRAM SHEET 2



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FORMULA SHEET

$\omega_d = \omega_n\sqrt{1-\zeta^2} = 2\pi f$ $\qquad t_p = \frac{1}{f}$

Number of oscillations $\frac{t_s}{t_p}$ or $\frac{2\sqrt{1-\zeta^2}}{\pi.\zeta}$

Damping coefficient $(\alpha) = \zeta.\omega_n = \frac{1}{\pi}\tau$

Overshoot $= e^{\frac{-\zeta\pi N}{\sqrt{1-\zeta^2}}}$

$\Psi = tan^{-1}\left[\frac{\sqrt{1-\zeta^2}}{-\zeta}\right] + \pi$ rad

Amplitude $= \varphi\left[1 + e^{\frac{-\zeta\pi N}{\sqrt{1-\zeta^2}}}\right]$

$\omega_n = \sqrt{\frac{Ko}{\tau}}$ $\qquad \tau + \frac{t_s}{4} = \frac{1}{\zeta.\omega_n}$

$\omega_p = \omega_n\sqrt{1-2\zeta^2}$

$\omega_b = \omega_n\sqrt{1-2\zeta^2+\sqrt{2-4\zeta^2+4\zeta^4}}$

$\frac{C}{R} = \frac{G}{1 \pm GH}$ $\qquad Sc = \frac{\sum P - \sum Z}{NP-NZ}$

$\zeta = \cos\varphi$ $\qquad \Psi = \frac{(2Ko+1)180^\circ}{NP-NZ}$

$Ko = \frac{\Delta P_1 . \Delta P_{2....}}{\Delta Z_1 . \Delta Z_{2....}}$

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AMPLIFIERS

$$\mathrm{Vo} = -V_i \frac{\mathrm{R}_f}{\mathrm{R}_1}$$

$$\mathrm{Vo} = V_i \left[\frac{\mathrm{R}_f}{\mathrm{R}_1}\right]$$

$$\mathrm{Vo} = -R_f \left[\frac{\mathrm{V}_1}{\mathrm{R}_1} + \frac{V_2}{R_2} + \cdots \frac{V_N}{R_N}\right]$$

$$\mathrm{Vo} = -\frac{1}{RC} \int V_i\,(t)dt + V_c$$

$$\mathrm{Vo} = -RC\,\frac{dV_i(t)}{dt}$$

$$\mathrm{i}_e = \frac{V_e}{R_e}$$

$$R_c = \frac{V_c}{i_c}$$

$$\mathrm{gm}R_L = \frac{hfe}{hie}.R_L$$

$$\mathrm{t} = \frac{1}{f}$$

$$\tau = \mathrm{R.C}$$

BODE AND NICHOLS CHARTS

$$\mathrm{Gain} = 20\log\left[\frac{\mathrm{output}}{\mathrm{input}}\right]\ \mathrm{dB}$$

$$\mathrm{Phase} = \sin\left[\frac{\mathrm{phase\ shift}}{\mathrm{input}}\right] - 180°$$

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LAPLACE TRANSFORM TABLE

No	F(s)	f(t)
1.	1	$\delta(t)$
2.	$\frac{A}{s}$	$A(t)$ $\{0 \;\; t < 0\}$ $\{A \;\; t \geq 0\}$
3.	$\frac{1}{s}$	$U(t)$ $\{0 \;\; t < 0\}$ $\{1 \;\; t \geq 0\}$
4.	$\frac{A}{s^2}$	At
5.	$\frac{2A}{s^3}$	At^3
6.	$\frac{A_\omega}{s^2 + \omega^2}$	$A \sin \omega t$
7.	$\frac{A_s}{s^2 + \omega^2}$	$A \cos \omega t$
8(a).	$\frac{A}{\tau s + 1}$	$\frac{A}{\tau} e^{\frac{-t}{\tau}}$
8(b).	$\frac{A}{s + a}$	Ae^{-at}
9(a).	$\frac{A}{(\tau_1 s + 1)(\tau_2 s + 1)}$	$\frac{A}{\tau_1 - \tau_2} \left[e^{\frac{-t}{\tau_1}} - \frac{-t}{\tau_2}\right]$
9(b).	$\frac{A}{(s + a)(s + b)}$	$\frac{A}{(b - a)} [e^{-at} - e^{-bt}]$
10(a).	$\frac{A}{(\tau_s + 1)^2}$	$\frac{At}{\tau^2} \left[e^{\frac{-t}{\tau}}\right]$
10(b).	$\frac{A}{(s + a)^2}$	Ate^{-at}
11.	$\frac{A{\omega_n}^2}{s^2 + 2\zeta\omega_n s + {\omega_n}^2}$	$\frac{A\omega_n e^{-\zeta\omega_n t}}{\sqrt{1 - \zeta^2}} \sin(\omega_n \sqrt{1 - \zeta^2 t}$
12(a).	$\frac{A}{s(\tau s + 1)}$	$A(1 - e^{\frac{-t}{\tau}})$
12(b).	$\frac{A}{s(s + a)}$	$\frac{A}{a}(1 - e^{-at})$
13(a).	$\frac{A}{s^2(\tau s + 1)}$	$A\tau \left[e^{\frac{-t}{\tau}} + \frac{t}{\tau} - 1\right]$
13(b).	$\frac{A}{s^2(s + a)}$	$\frac{A}{a^2} (e^{-at} + at - 1)$

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14(a).	$\frac{A\omega}{(s^2 + \omega^2)(\tau s + 1)}$	$\frac{A\omega\tau}{1 + \omega^2\tau^2} e^{\frac{-t}{\tau}} + \frac{A}{\sqrt{1 + \omega^2\tau^2}} \sin(\omega t - \Psi)$ where $\Psi = \tan^{-1}\omega\tau \ (0 < \Psi < \pi)$
14(b).	$\frac{A\omega}{(s^2 + \omega^2)(s + a)}$	$\frac{A\omega e^{-at}}{(\omega^2 + a^2)} + \frac{A}{\sqrt{\omega^2 + a^2}} \sin(\omega t - \Psi)$ where $\Psi = \tan^{-1}\frac{\omega}{a} \ (0 < \Psi < \pi)$
15(a).	$\frac{A}{s(\tau_1 s + a)(\tau_2 s + 1)}$	$A\left[1 + \frac{\tau_1 e^{\frac{-t}{\tau_1}} - \tau_2 e^{\frac{-t}{\tau_2}}}{\tau_1 - \tau_2}\right]$
15(b).	$\frac{A}{s(s + a)(s + b)}$	$\frac{A}{ab}\left[1 + \frac{ae^{-bt} - be^{-at}}{b - a}\right]$
16(a).	$\frac{A}{s(\tau + a)^2}$	$A\left[1 - \frac{(\tau + t)}{\tau} e^{\frac{-t}{\tau}}\right]$
16(b).	$\frac{A}{s(s + 1)^2}$	$\frac{A}{a^2}[1 - (1 + at)e^{-at}]$
17.	$\frac{A{\omega_n}^2}{s(s^2 + 2\zeta\omega_n s + {\omega_n}^2)}$	$A\left[1 + \frac{e^{-\zeta\omega_n t}}{\sqrt{1 - \zeta^2}} \sin(\omega_n t\sqrt{1 - \zeta^2} - \Psi\right]$ where $\Psi = \tan^{-1}\frac{\sqrt{1-\zeta^2}}{-\zeta} (0 < \Psi < \pi)$
18(a).	$\frac{A}{s^2(\tau_1 s + 1)(\tau_2 s + 1)}$	$A\left[t - \tau_1 - \tau_2 - \frac{{\tau_2}^2 e^{\frac{-t}{\tau_2}} - {\tau_1}^2 e^{\frac{-t}{\tau_1}}}{\tau_1 - \tau_2}\right]$
18(b).	$\frac{A}{s^2(s + a)(s + b)}$	$\frac{A}{ab}\left[t - \frac{a + b}{ab} - \frac{\frac{b}{a} e^{-bt} - \frac{a}{b} e^{-at}}{b - a}\right]$
19(a).	$\frac{A}{s^2(\tau s + 1)^2}$	$A\left[t - 2\tau + (t + 2\tau)e^{\frac{-t}{\tau}}\right]$
19(b).	$\frac{A}{s^2(s + a)^2}$	$\frac{A}{a^2}\left[t - \frac{2}{a} + \left(t + \frac{2}{a}\right)e^{-at}\right]$
20.	$\frac{A{\omega_n}^2}{s^2(s^2 + 2\zeta\omega_n s + {\omega_n}^2)}$	$A\left[\tau - \frac{2\zeta}{\omega_n} + \frac{e^{-\zeta\omega_n t}}{\omega_n\sqrt{1 - \zeta^2}} \sin(\omega_n t\sqrt{1 - \zeta^2} - \Psi\right]$ where $\Psi = 2\tan^{-1}\frac{\sqrt{1-\zeta^2}}{-\zeta} (0 < \Psi < \pi)$

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21(a).	$\frac{A\omega}{(s^2+\omega^2)(\tau_1 s+1)(\tau_2 s+1)}$	$A\left[\frac{{\tau_1}^2\omega e^{\frac{-t}{\tau_1}}}{(\tau_1-\tau_2)(1+\omega^2{\tau_1}^2)}+\frac{{\tau_2}^2\omega e^{\frac{-t}{\tau_2}}}{(\tau_1-\tau_2)(1+\omega^2{\tau_2}^2)}+\frac{\sin(\omega t-\Psi)}{(1+\omega^2\tau^2)(1+\omega^2{\tau_2}^2)^{\frac{1}{2}}}\right]$ where $\Psi=\tan^{-1}\omega\tau_1+\tan^{-1}\omega\tau_2$
21(b).	$\frac{A\omega}{(s^2+\omega^2)(s+a)(s+b)}$	$A\left[\frac{\omega e^{-at}}{(b-a)(\omega^2+a^2)}+\frac{\omega e^{-bt}}{(a-b)(\omega^2+b^2)}+\frac{\sin(\omega t-\Psi)}{(\omega^2+a^2)(\omega^2+b^2)^{\frac{1}{2}}}\right]$ where $\Psi=\tan^{-1}\frac{\omega(a+b)}{ab-\omega^2}\ (0<\Psi<\pi)$
22(a).	$\frac{A\omega}{(s^2+\omega^2)(\tau s+1)^2}$	$\frac{A}{1+\omega^2\tau^2}\left[\frac{\omega t+2\omega\tau}{1+\omega^2\tau^2}e^{\frac{-t}{\tau}}+\sin(\omega t-\Psi\right]$ where $\Psi=2\tan^{-1}\omega\tau$
22(b).	$\frac{A\omega}{(s^2+\omega^2)(s+a)^2}$	$\frac{A}{\omega^2+a^2}\left[\frac{a\omega(at+2)e^{-at}}{\omega^2+a^2}+\sin(\omega t-\Psi\right]$
23.	$\frac{A\omega{\omega_n}^2}{(s^2+\omega^2)(s^2+2\zeta\omega_n s+{\omega_n}^2)}$	$\frac{A{\omega_n}^2}{[({\omega_n}^2-\omega^2)^2+4\zeta^2\omega^2{\omega_n}^2]^{\frac{1}{2}}}\left[\sin(\omega t-\Psi)+\frac{\omega e^{-\zeta\omega_n t}\sin(\omega_n\sqrt{1-\zeta^2}\,t-\Psi_2)}{\omega_n\sqrt{1-\zeta^2}}\right]$ where $\Psi_1=tan^{-1}\left[\frac{2\zeta\omega\omega_n}{{\omega_n}^2+\omega^2}\right]\ (0<\Psi_1<\pi)$ and $\Psi_2=tan^{-1}-\frac{2\zeta{\omega_n}^2\sqrt{1-\zeta^2}}{\omega^2-{\omega_n}^2(1-2\zeta^2)}\ (0<\Psi_2<\pi)$

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